

TrES-1b

R-0250

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250629 · created 2026-07-17T06:26:44+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460855.909 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.016
Transit depth (Rp/Rs)^2	1.75 +/- 0.41 [%]
Orbital Inclination (inc)	85.54 +/- 0.24
Airmass coefficient 1 (a1)	1.003 +/- 0.011
Airmass coefficient 2 (a2)	0.001 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#2 - [562, 289]
Optimal Aperture	3.74
Optimal Annulus	11.26
Transit Duration (day)	0.0719 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.016 vs 0.1358 (deviation 0.21σ)
Duration fit vs published	0.0719 d vs 0.1042 d (deviation 5.8σ)
Mid-time O–C	-6.9 min
Depth SNR	7.3
Residual scatter	1.12 %

Light curve

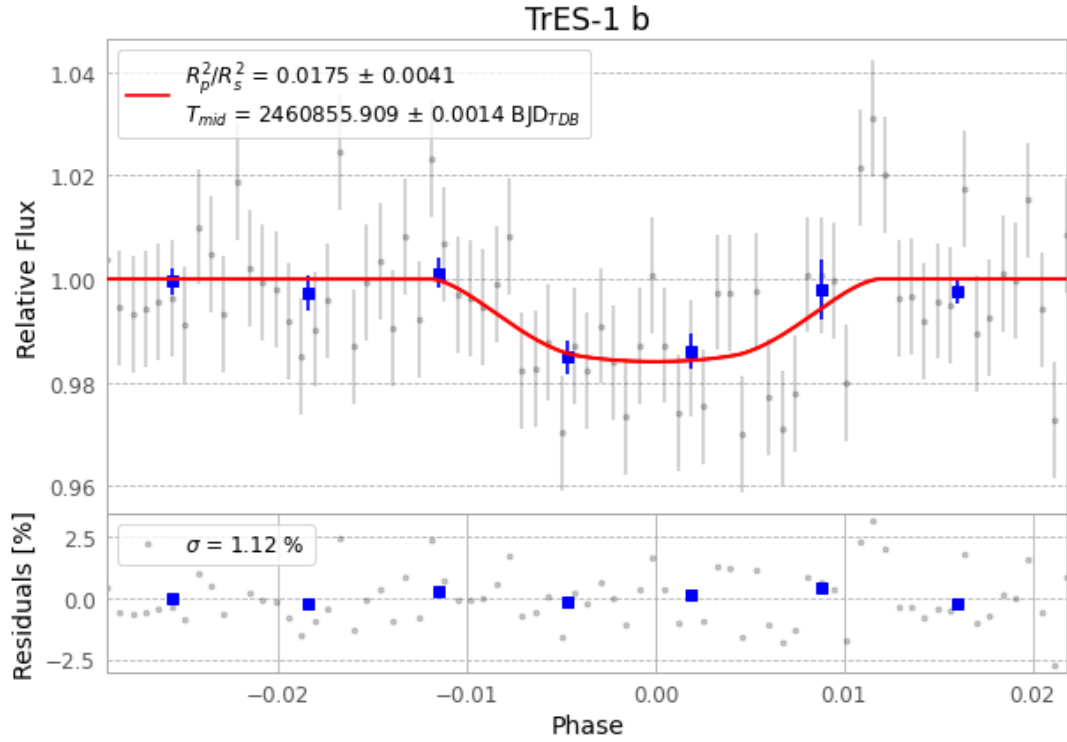


Figure 1 — FinalLightCurve_TrES-1 b_2025-06-29.png (SHA-256 27b070192ae60d21...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9f51b1d83b3720de...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

75 FITS frames (49 MB), TRES-1250629073616.FITS → TRES-1250629111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250629.log (867e8745e293...), FinalLightCurve_TrES-1 b_2025-06-29.png (27b070192ae6...), FinalLightCurve_TrES-1 b_2025-06-29.csv (d4ecda072280...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:26Z.